25 th April 2017.

$^3He + {}^{14}N$

1

^{14}N inelastic couplings.

We have $B(E\lambda)$ for all the states required from:
D. F. Geesaman et al., Phys. Rev. C27, 1134 (1983).

a) 0.0 1^+ $\longrightarrow$ 3.95 1^+ $B(E2)\uparrow = 2.78\ e^2 fm^4$

b) " $\longrightarrow$ 5.11 2^- $B(E3)\uparrow = 74\ e^2 fm^6$

c) " $\longrightarrow$ 5.83 3^- $B(E3)\uparrow = 117\ e^2 fm^6$

d) " $\longrightarrow$ 7.03 2^+ $B(E2)\uparrow = 3.95\ e^2 fm^4$

These must be converted into the form used by FRESCO.
See pp. 14 - 15 of manual.

We will use couplings of TYPE = 13 (target excitation).
We need M (Ek)

Take case (a) first:

$$M(E2) = \pm\sqrt{(2\times 1+1)\times 2.78}$$

$$= \pm\sqrt{3 \times 2.78}$$

$$= \pm\sqrt{8.34}$$

$$= \underline{\underline{\pm\ 2.8879\ e\ fm^2}}$$

This is for $1^+ \rightarrow 1^+$ (i.e. $1 \rightarrow 2$ in terms of states in FRESCO input)

Same for $2 \rightarrow 1$.

If we want reorientation according to collective model we can also add this if required. For the moment we will omit it.